

IGCSE • Cambridge (CIE) • Maths

 2 hours  31 questions

Target Test

Created 29/05/26

Total Marks

/123

Scan here for the answers
or visit [savemyexams.com](https://www.savemyexams.com)



- 1 (a)** The coordinates of P are $(-3, 8)$ and the coordinates of Q are $(9, -2)$.

Find the equation of the perpendicular bisector of PQ .

(4 marks)

- (b)** The coordinates of P are $(-3, 8)$ and the coordinates of Q are $(9, -2)$.

Find the equation of the line parallel to PQ that passes through the point $(6, -1)$.

(3 marks)

- 2** Show that the line $4y = 5x - 10$ is perpendicular to the line $5y + 4x = 35$.

(3 marks)

- 3** A straight line goes through the points (p, q) and (r, s) , where

- $p + 2 = r$
- $q + 4 = s$.

Find the gradient of the line.

(3 marks)

- 4 (a) A is the point $(8,5)$ and B is the point $(-4,1)$.

Calculate the length of AB .

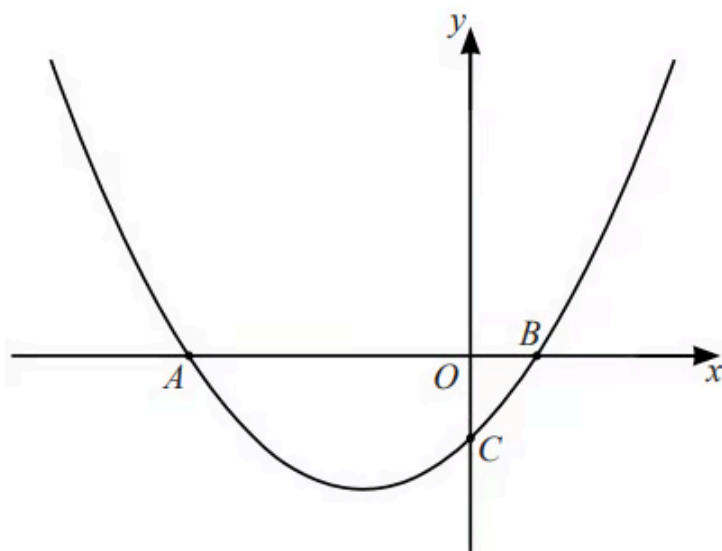
(3 marks)

- (b) Find the co-ordinates of the midpoint of AB .

(..... ,)

(2 marks)

5



NOT TO
SCALE

The diagram shows a sketch of the curve $y = x^2 + 3x - 4$.

Find the coordinates of the points A , B and C .

A (..... ,)

B (..... ,)

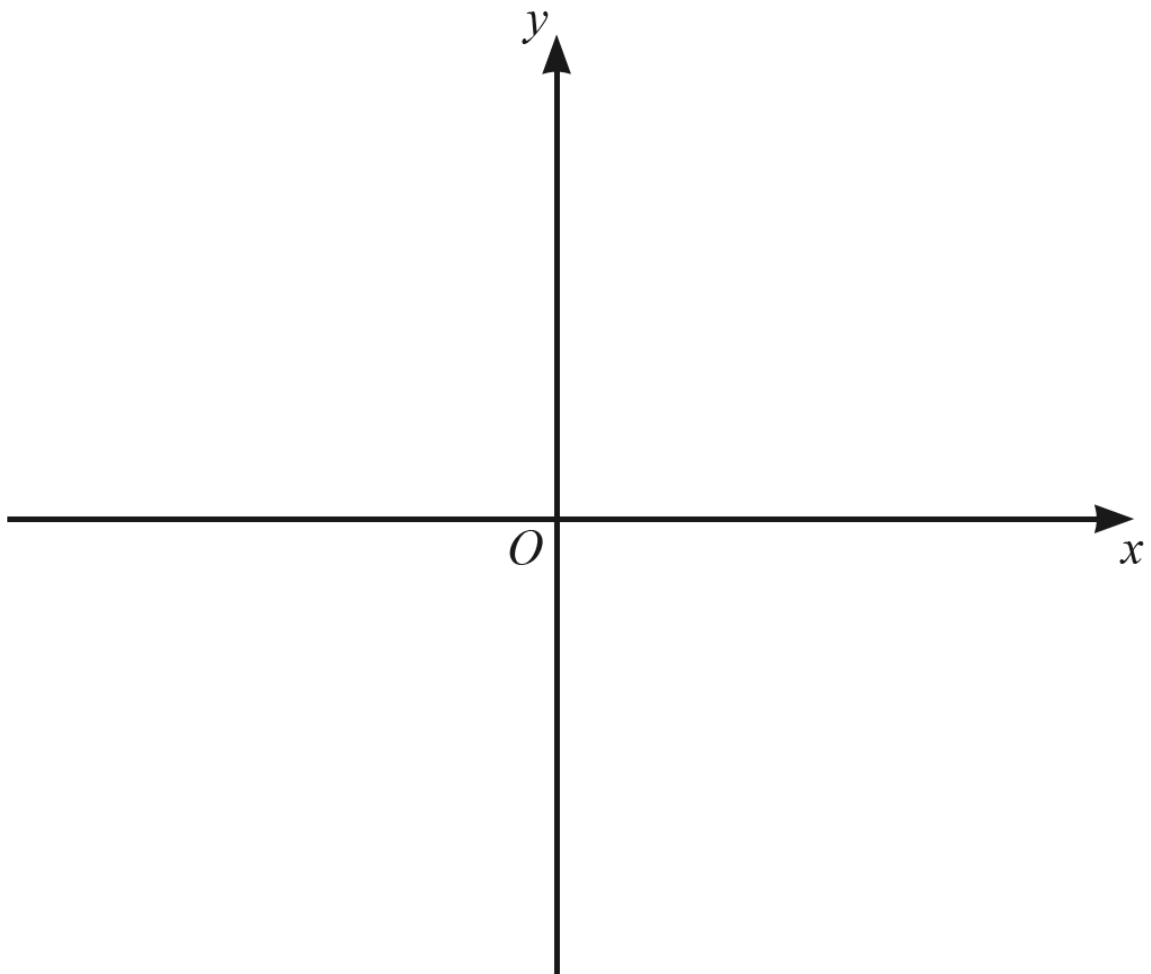
C (.....,)

(4 marks)

6 (a) Write $x^2 + 10x + 14$ in the form $(x + a)^2 + b$.

(2 marks)

(b) Sketch the graph of $y = x^2 + 10x + 14$, indicating the coordinates of the turning point.

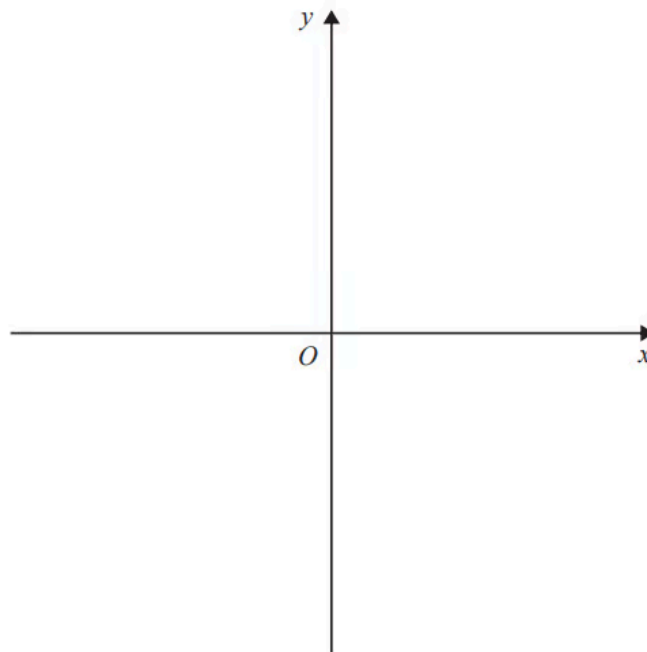


(3 marks)

7 The curve C has equation $y = x^2 - 6x + 4$

Using the axes below, sketch the curve C . On your sketch show clearly

- i) the exact coordinates of any points of intersection of C with the coordinate axes,
- ii) the coordinates of the turning point.



(6 marks)

8 (a) The point P has co-ordinates $(10, 12)$ and the point Q has co-ordinates $(2, -4)$. Find

- i) the co-ordinates of the mid-point of the line PQ ,

(.....,) [2]

- ii) the gradient of the line PQ ,

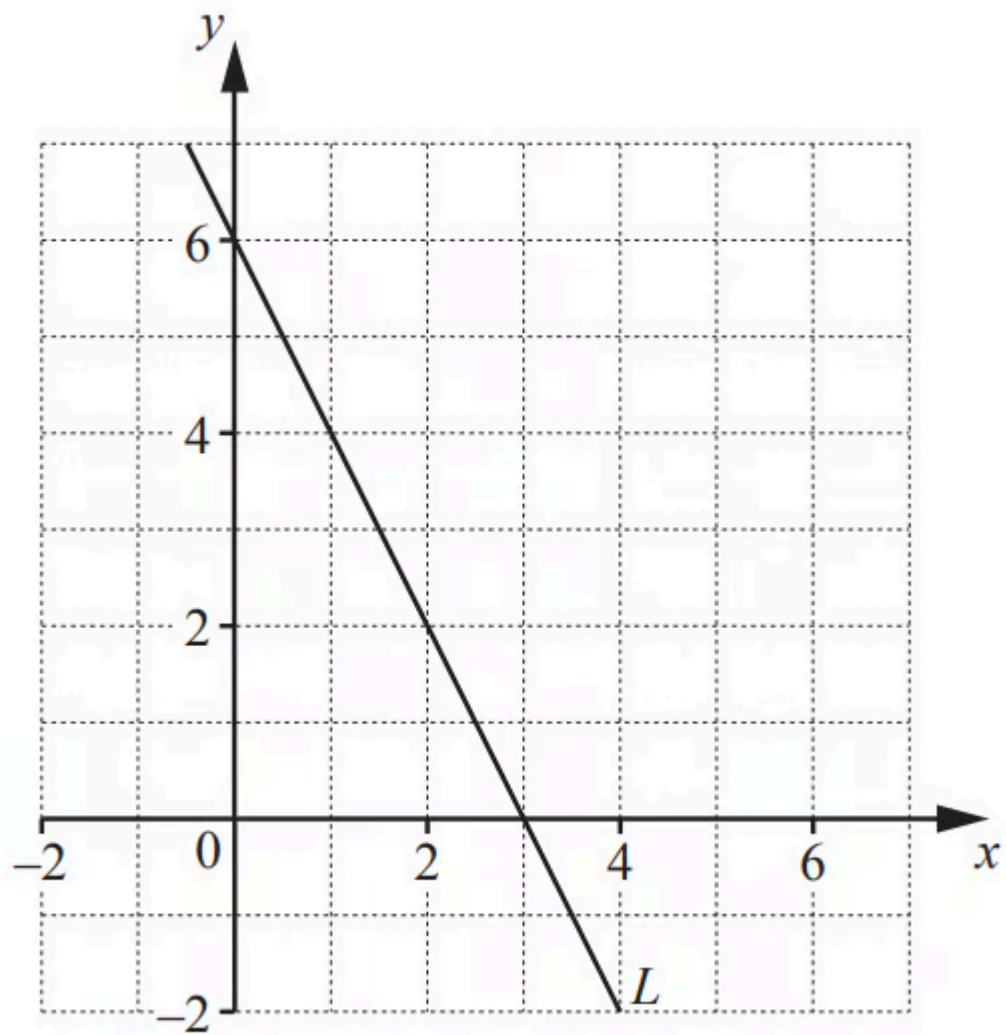
[2]

(4 marks)

(b) Find the equation of a line perpendicular to PQ that passes through the point $(2, 3)$.

(3 marks)

9 (a) The diagram shows a straight line L .



Find the equation of line L .

(3 marks)

(b) Find the equation of the line perpendicular to line L that passes through $(9, 3)$.

(3 marks)

- 10** A straight line passes through the points $(45, 130)$ and $(95, 170)$.

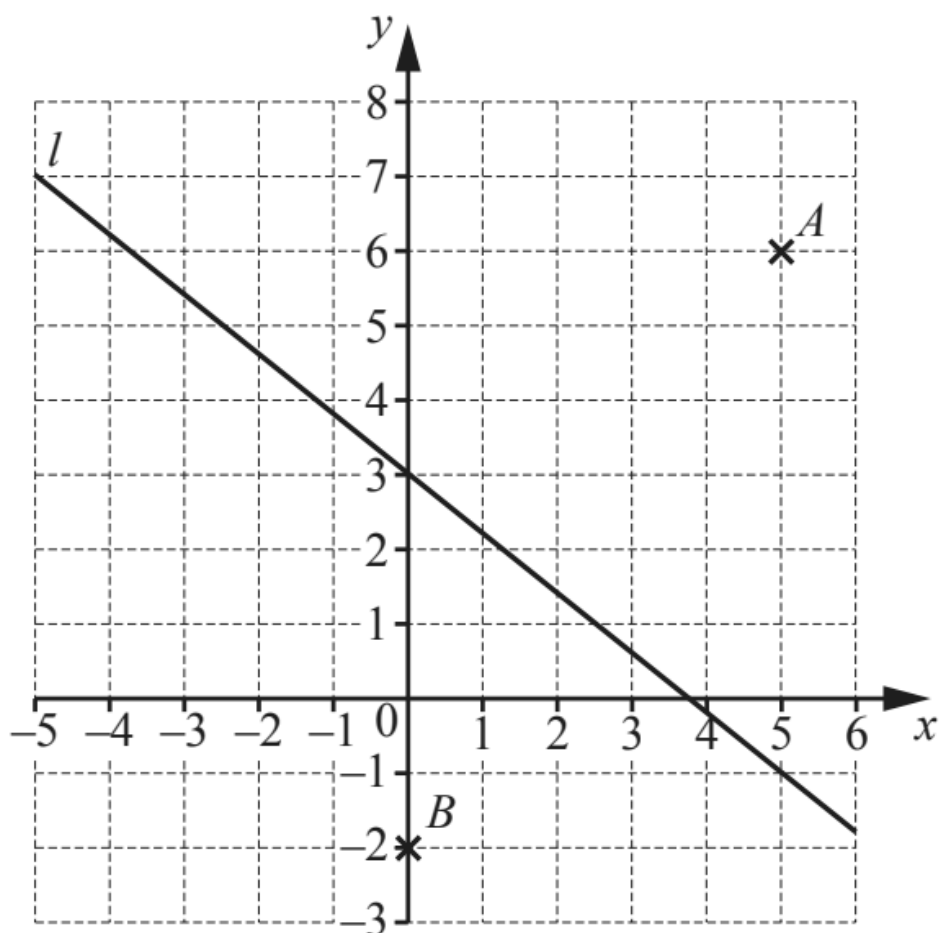
Find the equation of the line in the form $y = mx + c$.

(4 marks)

- 11** Point A has coordinates $(5, 8)$
Point B has coordinates $(9, -4)$

Work out the gradient of AB .

(2 marks)



C is the point $(8, 14)$.

Write down the equation of the line perpendicular to line l that passes through the point C.

(3 marks)

13 Line L passes through the points $(0, -3)$ and $(6, 9)$.

Find the equation of line L .

(3 marks)

- 14** Find the equation of the straight line that is perpendicular to the line $y = \frac{1}{2}x + 1$ and passes through the point $(1, 3)$.

(3 marks)

- 15** Find the equation of the line that is perpendicular to the line $y = 2x - 3$ and passes through the point $(0, 2)$.

(2 marks)

- 16** A is the point $(7, 2)$ and B is the point $(-5, 8)$.

Calculate the length of AB .

(3 marks)

- 17 (a)** A rhombus $ABCD$ has a diagonal AC where A is the point $(-3, 10)$ and C is the point $(4, -4)$.

i) Calculate the length AC .

[3]

ii) Show that the equation of the line AC is $y = -2x + 4$.

[2]

(5 marks)

(b) Find the equation of the line BD .

(4 marks)

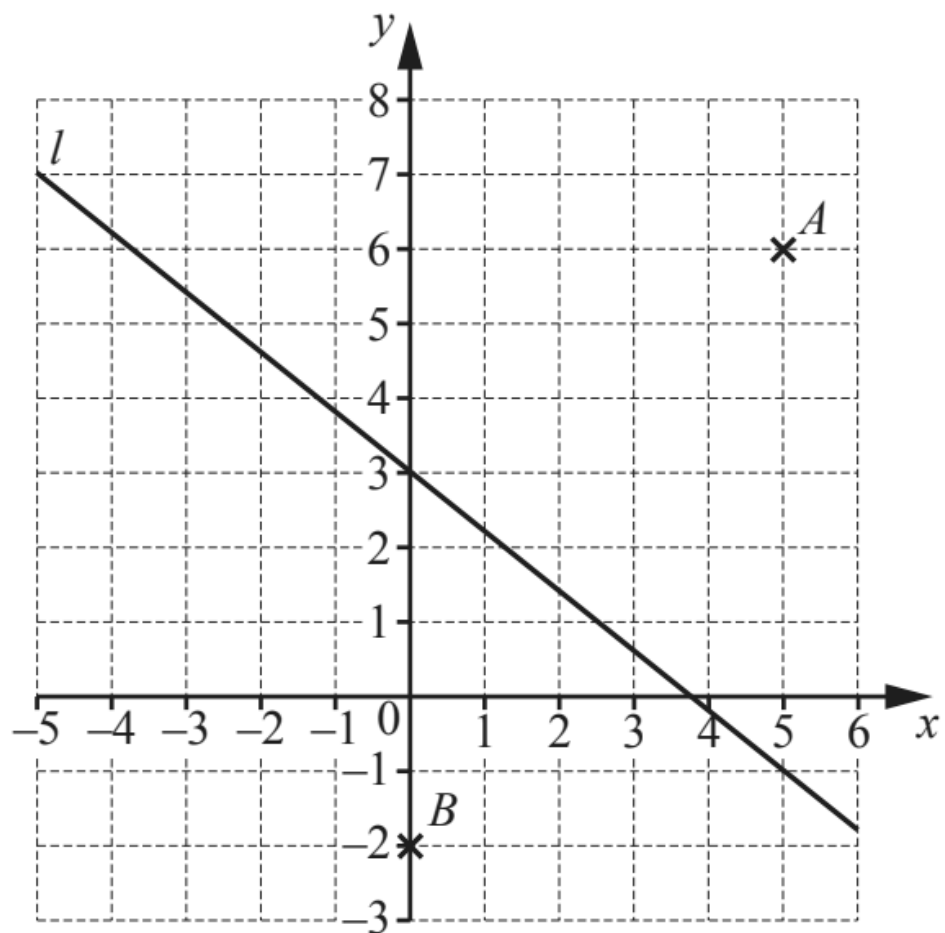
18 A straight line joins the points $(3k, 6)$ and $(k, -5)$. The line has a gradient of 2.

Find the value of k .

$k = \dots\dots\dots$

(3 marks)

19 (a)



C is the point $(8, 14)$.

Calculate the length of AC .

(3 marks)

(b) Find the co-ordinates of the mid-point of BC .

(..... ,)

(2 marks)

- 20** A line joins $A(1, 3)$ to $B(5, 8)$.

Find the equation of the line AB .

Give your answer in the form $y = mx + c$.

$y = \dots\dots\dots$

(3 marks)

- 21** A rhombus $ABCD$ has a diagonal AC where A is the point $(-3, 10)$ and C is the point $(4, -4)$.

Calculate the length AC .

(3 marks)

- 22** Find the gradient of a line that is perpendicular to $8y + 4x = 5$.

(2 marks)

- 23** A is the point $(7, 12)$ and B is the point $(2, -1)$.

Find the length of AB .

(3 marks)

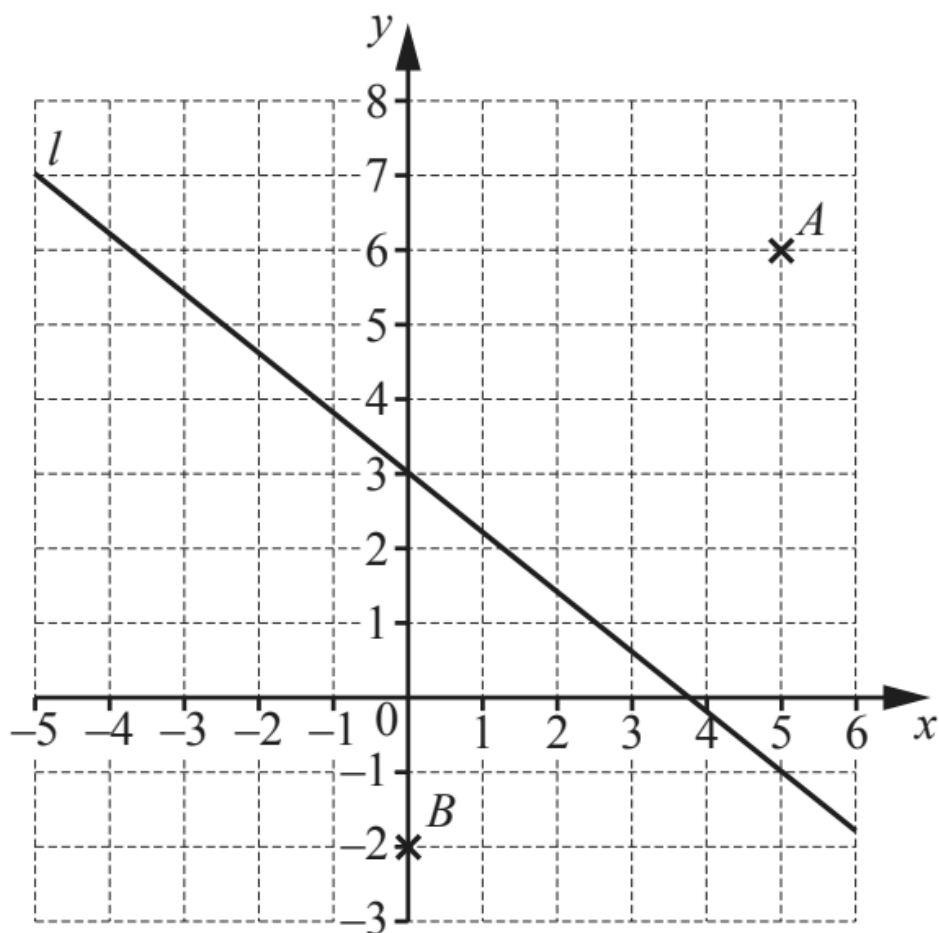
- 24 The line PQ has equation $y = 3x - 8$ and point P has co-ordinates $(6, 10)$.

Find the equation of the line that passes through P and is perpendicular to PQ .
Give your answer in the form $y = mx + c$.

$y = \dots\dots\dots$

(3 marks)

25



Write down the equation of the line parallel to line l that passes through the point B .

(2 marks)

26 The points A , B and C lie in order on a straight line.

The coordinates of A are $(2, 5)$

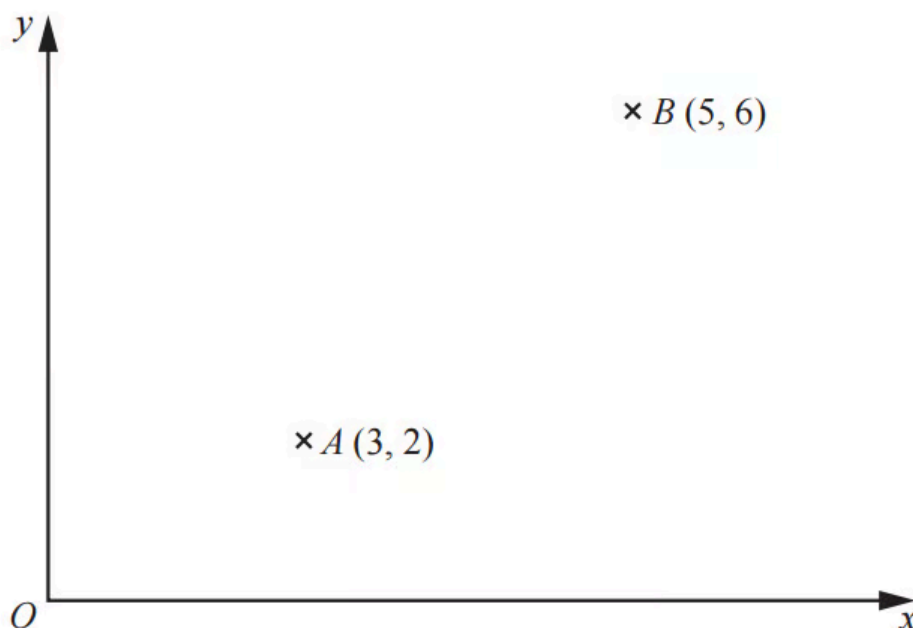
The coordinates of B are $(4, p)$

The coordinates of C are $(q, 17)$

Given that $AC = 4AB$, find the values of p and q .

(3 marks)

27 (a)



NOT TO
SCALE

B is the mid-point of the line AC .

Find the co-ordinates of C .

(..... ,)

(2 marks)

- (b) Find the equation of the straight line that passes through A and B .

(3 marks)

- (c) The straight line that passes through A and B cuts the y -axis at D .

Write down the co-ordinates of D .

(..... ,)

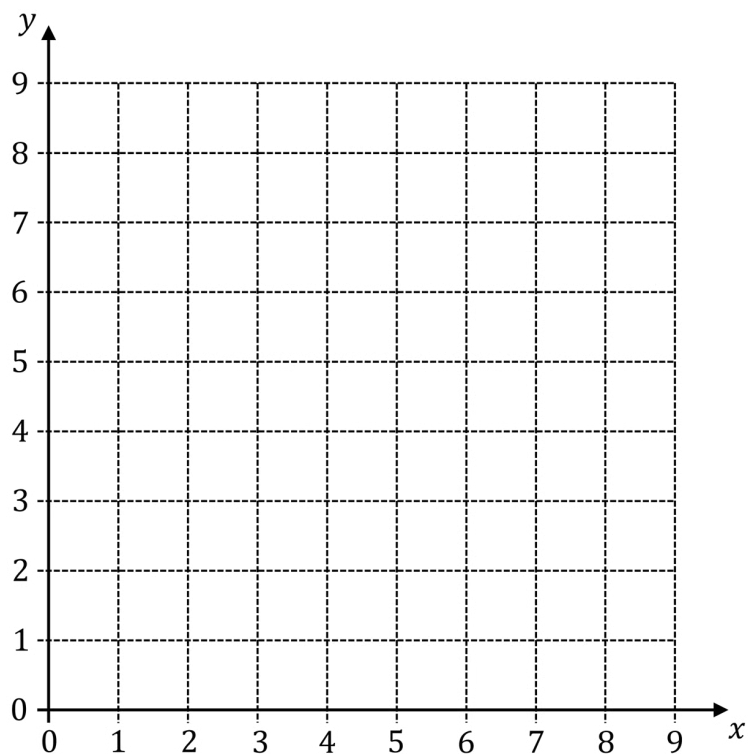
(1 mark)

- 28 P is the point $(16, 9)$ and Q is the point $(22, 24)$.

Find the equation of the line perpendicular to PQ that passes through the point $(5, 1)$.
Give your answer in the form $y = mx + c$.

$y =$

(4 marks)



On the grid, draw the lines $y = x$ and $x + y = 8$.

(3 marks)

30 A triangle has vertices P , Q and R .

The coordinates of P are $(-3, -6)$

The coordinates of Q are $(1, 4)$

The coordinates of R are $(5, -2)$

M is the midpoint of PQ .

N is the midpoint of QR .

Prove that MN is parallel to PR .

You must show each stage of your working.

(4 marks)

31 A is the point $(7, 2)$ and B is the point $(-5, 8)$.

Find the equation of the line that is perpendicular to AB and that passes through the point $(-1, 3)$.

Give your answer in the form $y = mx + c$.

$y = \dots\dots\dots$

(4 marks)